

Figure TF17-2 Different types of electromagnetic radiation.

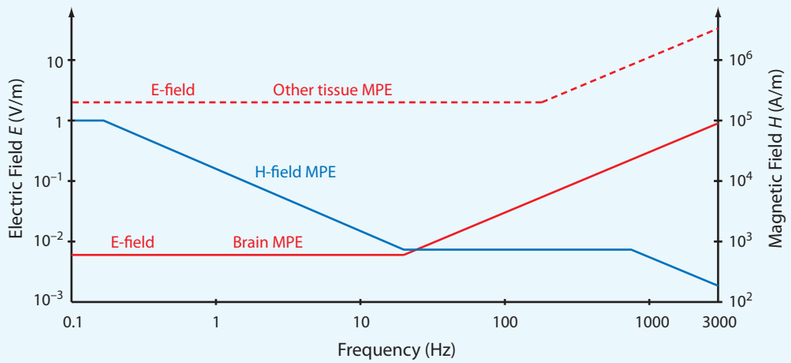


Figure TF17-3 Maximum permissible exposure (MPE) levels for E and H over the frequency range from 0.1 Hz to 3 kHz.

Frequency Range $0 \le f \le 3$ kHz: The plots in **Fig. TF17-3** display the values of MPE for electric and magnetic fields over the frequency range below 3 kHz. According to IEEE Std C95.6, it is sufficient to demonstrate compliance with the MPE levels for either the electric field E or the magnetic field H. According to the plot for H, exposure at 60 Hz should not exceed 720 A/m. The magnetic field due to power lines is typically in the range of 2 to 6 A/m underneath the lines, which is at least two orders of magnitude smaller than the established safe level for H.

Frequency Range 3 kHz $\le f \le 300$ GHz: At frequencies below 500 MHz, MPE is specified in terms of the electric and magnetic field strengths of the EM energy (**Fig. TF17-4**). From 100 MHz to 300 GHz (and beyond), MPE is specified in terms of the product of E and H, namely the power density S. Cell phones operate in the 1 to 2 GHz band; the specified MPE is 1 W/m^2 (or equivalently 0.1 mW/cm^2).

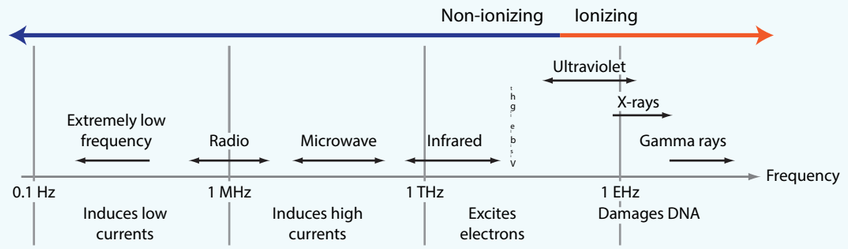


图 TF17-2 不同类型的电磁辐射。

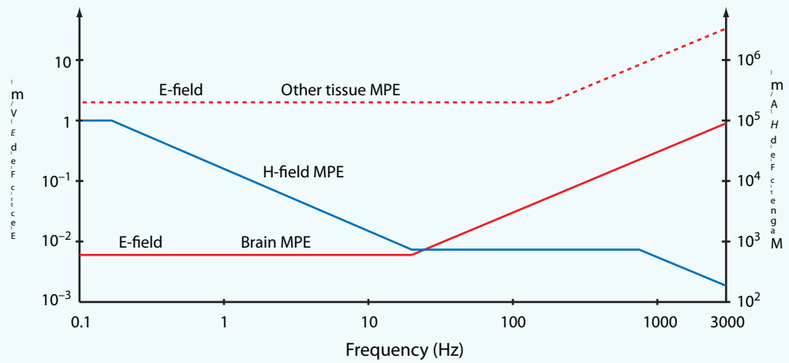


图 TF17-3 E 和 H 在 0.1 Hz 至 3 kHz 频率范围内的最大允许暴露 (MPE) 水平。

频率范围 $0 \le f \le 3$ kHz：图 TF17-3 中的图显示了 3 kHz 以下频率范围内电场和磁场的 MPE 值。根据 IEEE 标准 C95.6，足以证明电场 E 或磁场 H 符合 MPE 水平。根据 H 绘图，60 Hz 下的暴露不应超过 720 A/m。电源线产生的磁场通常在线路下方 2 至 6 A/m 的范围内，这比既定的 H 安全水平至少小两个数量级。

频率范围 3 kHz $\le f \le$ 300 GHz：在低于 500 MHz 的频率下，MPE 根据 EM 能量的电场和磁场强度来指定（图 TF 17-4）。从 100 MHz 到 300 GHz（及以上），MPE 以 E 和 H 的乘积来指定，即功率密度 S。手机工作在 1 到 2 G Hz 频段；指定的 MPE 为 1 W/m^2（或等效的 0.1 mW/cm ^{2}）。

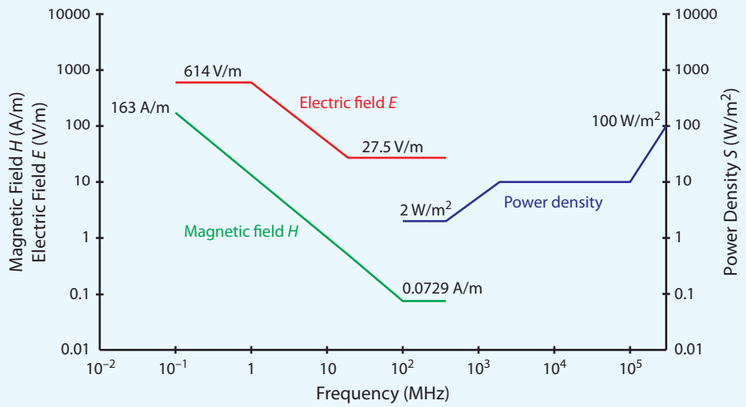


Figure TF17-4 MPE levels for the frequency range from 10 kHz to 300 GHz.

Bottom Line

We are constantly bombarded by EM energy—from solar illumination to blackbody radiation emitted by all matter. Our bodies absorb, reflect, and emit EM energy all the time. Living organisms, including humans, require exposure to EM radiation to survive, but excessive exposure can cause adverse effects. The term *excessive exposure* connotes a complicated set of relationships among such variables as field strength, exposure duration and mode (continuous, pulsed, etc.), body part, etc. The emission standards established by the Federal Communications Commission in the U.S. and similar governmental bodies in other countries are based on a combination of epidemiological studies, experimental observations, and theoretical understanding of how EM energy interacts with biological material. Generally speaking, the maximum permissible exposure levels specified by these standards are typically two orders of magnitude lower than the levels known to cause adverse effects, but in view of the multiplicity of variables involved, there is no guarantee that adhering to the standards will avoid health risks absolutely. The bottom line is: Use common sense!

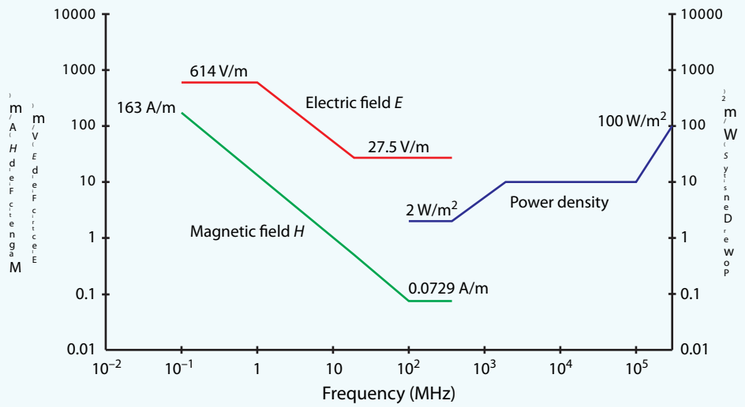


图 TF17-4 10 kHz 至 300 GHz 频率范围的 MPE 电平。

底线

我们不断受到电磁能量的轰炸——从太阳光到所有物质发出的黑体辐射。我们的身体一直在吸收、反射和发射电磁能量。包括人类在内的生物体需要暴露于电磁辐射才能生存，但过度暴露会造成不利影响。术语“过度暴露”意味着诸如场强、暴露持续时间和模式（连续、脉冲等）、身体部位等变量之间的一组复杂关系。美国联邦通信委员会和其他国家类似政府机构制定的标准基于流行病学研究、实验观察以及对电磁能如何与生物材料相互作用的理论理解。一般来说，这些标准规定的最大允许暴露水平通常比已知会造成不利影响的水平低两个数量级，但考虑到涉及的变量的多样性，不能保证遵守这些标准就绝对避免健康风险。底线是：运用常识!

Exercise 9-10: The effective area of a parabolic dish antenna is approximately equal to its physical aperture. If its directivity is 30 dB at 10 GHz, what is its effective area? If the frequency is increased to 30 GHz, what will be its new directivity?

Answer: $A_e = 0.07\ m^2$, $D = 39.44$ dB. (See EM.)

9-7 Radiation by Large-Aperture Antennas

For wire antennas, the sources of radiation are the infinitesimal current elements comprising the current distribution along the wire and the total radiated field at a given point in space is equal to the sum, or integral, of the fields radiated by all the elements. A parallel scenario applies to aperture antennas, except that now the source of radiation is the electric-field distribution across the aperture. Consider the horn antenna shown in **Fig. 9-20**. It is connected to a source through a coaxial transmission line with the outer conductor of the line connected to the metal body of the horn and the inner conductor made to protrude through a small hole partially into the throat end of the horn. The protruding conductor acts as a monopole antenna, generating waves that radiate outwardly toward the horn's aperture. The electric field of the wave arriving at the aperture, which may vary as a function of x_a and y_a over the horn's aperture, is called the ***electric-field aperture distribution*** or ***illumination***, $E_a(x_a, y_a)$. Inside the horn, wave propagation is guided by the horn's geometry, but as the wave transitions from a guided wave into an unbounded wave, every point of its wavefront serves as a source of spherical secondary wavelets. The aperture may then be represented as a distribution of isotropic radiators. At a distant point Q, the combination of all the waves arriving from all of these radiators constitutes the total wave that would be observed by a receiver placed at that point.

The radiation process described for the horn antenna is equally applicable to any aperture upon which an electromagnetic wave is incident. For example, if a light source is used to illuminate an opening in an opaque screen through a collimating lens, as shown in **Fig. 9-21(a)**, the opening becomes a source of secondary spherical wavelets, much like the aperture of the horn antenna. In the case of the parabolic reflector shown in **Fig. 9-21(b)**, it can be described in terms of an imaginary aperture representing the electric-field distribution across a

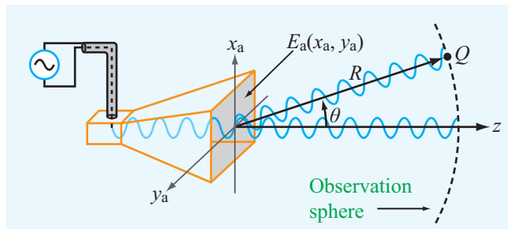


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

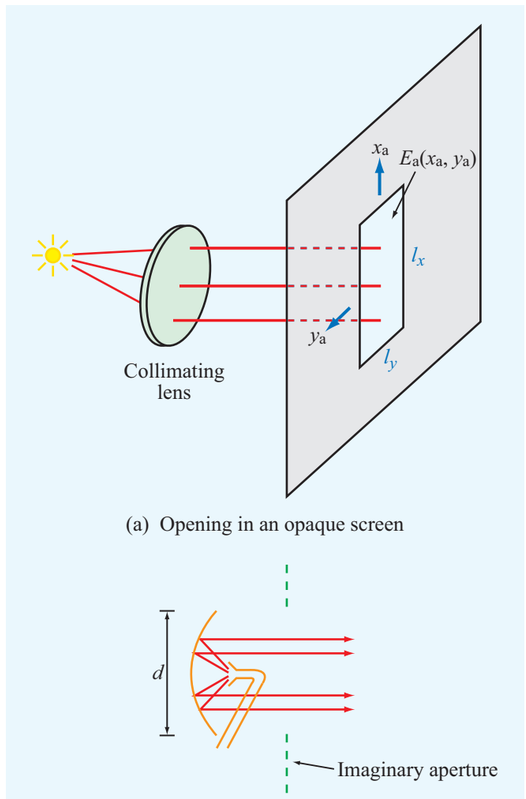


Figure 9-21 Radiation by apertures: (a) an opening in an opaque screen illuminated by a light source through a collimating lens and (b) a parabolic dish reflector illuminated by a small horn antenna.

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练习9-10：抛物面碟形天线的有效面积大约等于其物理孔径。如果它的方向性在 10 GHz 时为 30 dB，那么它的有效面积是多少？如果频率提高到30GHz，它新的指向性会是什么？

答案：$A_e = 0.07\ m^2$，D = 39.44 dB。 （参见EM。）

9-7 大孔径天线的辐射

对于线天线，辐射源是由沿线的电流分布组成的无穷小电流元件，空间中给定点的总辐射场等于所有元件辐射场的总和或积分。类似的情况适用于孔径天线，只不过现在的辐射源是孔径上的电场分布。考虑图 9-20 所示的喇叭天线。它通过同轴传输线连接到信号源，该线的外导体连接到喇叭的金属主体，而内导体则通过小孔部分突出到喇叭的喉部末端。突出的导体充当单极天线，产生向喇叭孔径向外辐射的波。到达孔径的波的电场可能随着喇叭孔径上的 x_a 和 y_a 的函数而变化，称为电场孔径分布或照明，$E_a(x_a, y_a)$。在喇叭内部，波传播由喇叭的几何形状引导，但当波从导波转变为无界波时，其波前的每个点都充当球形次级小波的源。然后孔径可以被表示为各向同性辐射器的分布。在远处的点 Q，来自所有这些辐射器的所有波的组合构成了通过以下方式观测到的总波：

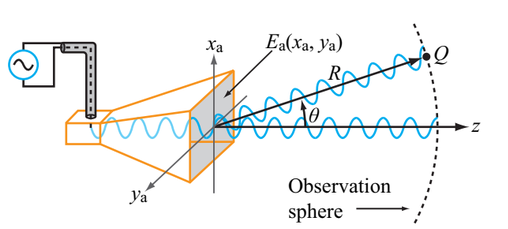


图 9-20 孔径场分布为 $E_a(x_a,y_a)$ 的喇叭天线。

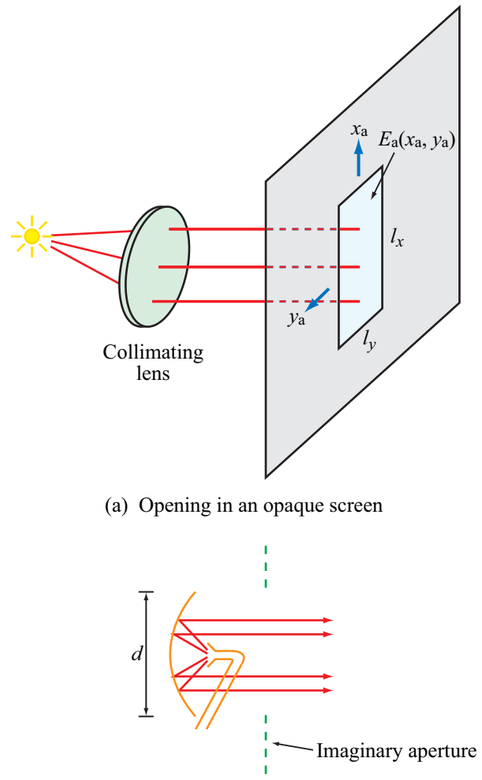


图 9-21 孔径辐射：(a) 不透明屏幕上的开口，通过准直透镜由光源照明；(b) 由小喇叭天线照明的抛物面碟形反射器。

针对喇叭天线描述的辐射过程同样适用于电磁波入射的任何孔径。例如，如果使用光源通过准直透镜照亮不透明屏幕上的开口，如图9-21（a）所示，则该开口就成为二次球面小波的源，就像喇叭天线的孔径一样。在图 9-21(b) 所示的抛物面反射器的情况下，它可以用代表电场分布的假想孔径来描述。

plane in front of the reflector.

Two types of mathematical formulations are available for computing the electromagnetic fields of waves radiated by apertures. The first is a ***scalar formulation*** based on Kirchhoff's work, and the second is a ***vector formulation*** based on Maxwell's equations. In this section, we limit our presentation to the scalar diffraction technique in part because of its inherent simplicity and also because it is applicable across a wide range of practical applications.

► The key requirement for the validity of the scalar formulation is that the antenna aperture be at least several wavelengths long along each of its principal dimensions. ◄

A distinctive feature of such an antenna is its high directivity and correspondingly narrow beam, which makes it attractive for radar and free-space microwave communication systems. The frequency range commonly used for such applications is the 1 to 30 GHz microwave band. Because the corresponding wavelength range is 30 to 1 cm, respectively, it is quite practical to construct and use antennas (in this frequency range) with aperture dimensions that are many wavelengths in size.

The x_a–y_a plane in **Fig. 9-22** (denoted plane A) contains an aperture with an electric field distribution $E_a(x_a, y_a)$. For the sake of convenience, the opening has been chosen to be rectangular in shape, with dimensions l_x along x_a and l_y along y_a, even though the formulation we are about to discuss is general enough to accommodate any two-dimensional aperture distribution, including those associated with circular and elliptical apertures. At a distance z from the aperture plane A in **Fig. 9-22**, we have an observation plane O with axes (x, y). The two planes have parallel axes and are separated by a distance z. Moreover, z is sufficiently large that any point Q in the observation plane is in the far-field region of the aperture. To satisfy the far-field condition, it is necessary that

$$R \geq 2d^2/\lambda, \qquad \textbf{(far-field range)} \tag{9.73}$$

where d is the longest linear dimension of the radiating aperture.

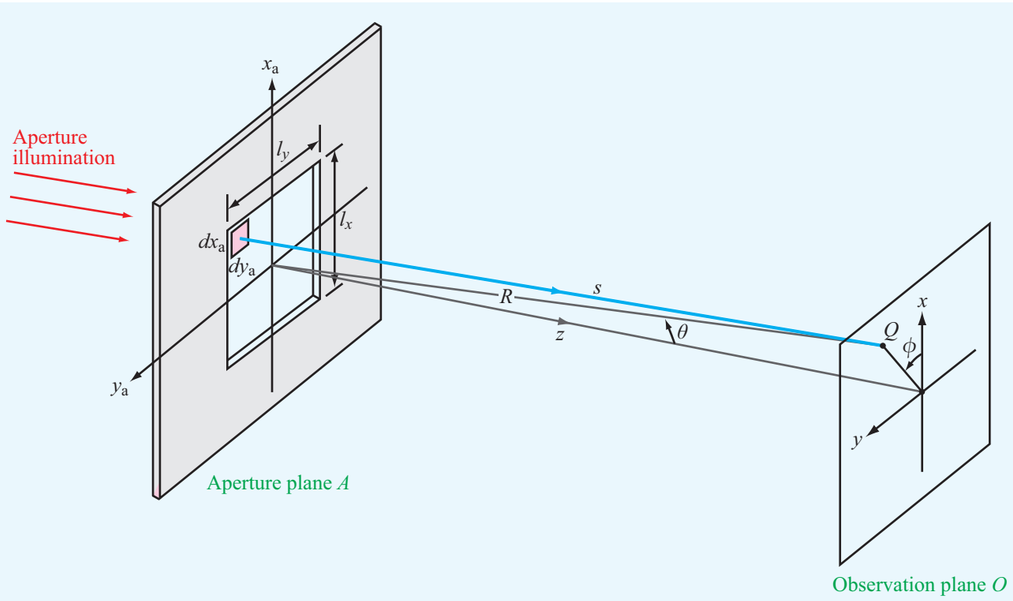


Figure 9-22 Radiation by an aperture in the x_a–y_a plane at $z = 0$.

平面位于反射镜前面。

有两种类型的数学公式可用于计算孔径辐射波的电磁场。第一个是基于基尔霍夫工作的标量公式，第二个是基于麦克斯韦方程组的矢量公式。在本节中，我们将介绍限制于标量衍射技术，部分原因是其固有的简单性，也因为它适用于广泛的实际应用。

► 标量公式有效性的关键要求是天线孔径沿其每个主要维度至少有几个波长长。 ◄

这种天线的一个显着特点是其高方向性和相应的窄波束，这使其对雷达和自由空间微波通信系统具有吸引力。此类应用通常使用的频率范围是 1 至 30 GHz 微波频段。因为对应的波长范围分别是30到1cm，所以相当

构造和使用孔径尺寸为多个波长的天线（在此频率范围内）是可行的。

图 9-22 中的 x_a–y_a 平面（表示为平面 A）包含电场分布为 $E_a(x_a、y_a)$ 的孔径。为了方便起见，开口形状选择为矩形，尺寸为沿 x_a 的 l_x 和沿 y_a 的 l_y，尽管我们将要讨论的公式足够通用，足以适应任何二维孔径分布，包括与圆形和椭圆形孔径相关的分布。在图 9-22 中，距孔径平面 A 的距离为 z 处，我们有一个轴为 (x, y) 的观察平面 O。这两个平面具有平行轴并且相距距离 z。此外，z 足够大，使得观察平面中的任何点 Q 都位于孔径的远场区域。为了满足远场条件，需要

$$R \geq 2d^2/\lambda, \qquad \textbf{(far-field range)} \tag{9.73}$$

其中 d 是辐射孔径的最长线性尺寸。

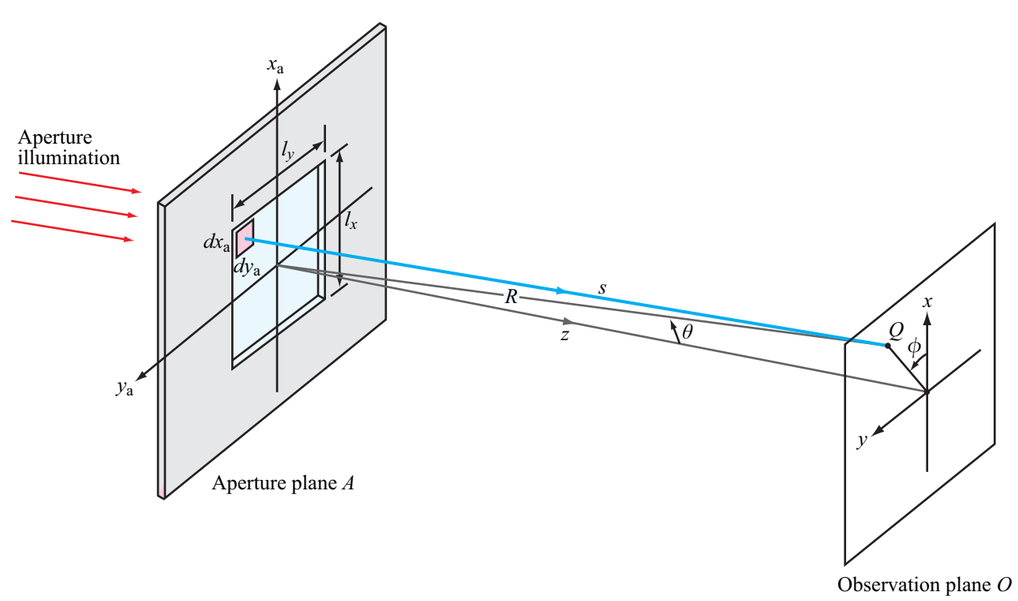


图 9-22 x_a–y_a 平面中 $z = 0$ 处孔径的辐射。

The position of observation point Q is specified by the range R between the center of the aperture and point Q and by the angles θ and ϕ (**Fig. 9-22**), which jointly define the direction of the observation point relative to the coordinate system of the aperture. In our treatment of the dipole antenna, we oriented the dipole along the z axis, and we called θ the zenith angle. In the present context, the z axis is orthogonal to the plane containing the antenna aperture. Also, θ usually is called the ***elevation angle***. The electric field phasor of the wave incident upon point Q is denoted $\widetilde{E}(R,\theta,\phi)$. Kirchhoff's scalar diffraction theory provides the following relationship between the radiated field $\widetilde{E}(R,\theta,\phi)$ and the aperture illumination $\widetilde{E}_{\mathrm{a}}(x_{\mathrm{a}},y_{\mathrm{a}})$:

$$\widetilde{E}(R,\theta,\phi)=\frac{j}{\lambda}\left(\frac{e^{-jkR}}{R}\right)\widetilde{h}(\theta,\phi), \tag{9.74}$$

where

$$\widetilde{h}(\theta,\phi)=\iint_{-\infty}^{\infty}\widetilde{E}_{\mathrm{a}}(x_{\mathrm{a}},y_{\mathrm{a}}) \cdot\exp[jk\sin\theta(x_{\mathrm{a}}\cos\phi+y_{\mathrm{a}}\sin\phi)]\,dx_{\mathrm{a}}\,dy_{\mathrm{a}}. \tag{9.75}$$

We shall refer to $\widetilde{h}(\theta,\phi)$ as the ***form factor*** of $\widetilde{E}(R,\theta,\phi)$. Its integral is written with infinite limits with the understanding that $\widetilde{E}_{\mathrm{a}}(x_{\mathrm{a}},y_{\mathrm{a}})$ is identically zero outside the aperture. The spherical propagation factor (e^{-jkR}/R) accounts for wave propagation between the center of the aperture and the observation point, and $\widetilde{h}(\theta,\phi)$ represents an integration of the exciting field $\widetilde{E}_{\mathrm{a}}(x_{\mathrm{a}},y_{\mathrm{a}})$ over the extent of the aperture, taking into account [through the exponential function in Eq. (9.75)] the approximate deviation in distance between R and s, where s is the distance to any point $(x_{\mathrm{a}},y_{\mathrm{a}})$ in the aperture plane (see **Fig. 9-22**).

► In Kirchhoff's scalar formulation, the polarization direction of the radiated field $\widetilde{E}(R,\theta,\phi)$ is the same as that of the aperture field $\widetilde{E}_{\mathrm{a}}(x_{\mathrm{a}},y_{\mathrm{a}})$. ◄

Also, the power density of the radiated wave is given by

$$S(R,\theta,\phi)=\frac{|\widetilde{E}(R,\theta,\phi)|^2}{2\eta_0}=\frac{|\widetilde{h}(\theta,\phi)|^2}{2\eta_0\lambda^2R^2}. \tag{9.76}$$

9-8 Rectangular Aperture with Uniform Aperture Distribution

To illustrate the scalar diffraction technique, consider a rectangular aperture of height l_x and width l_y, both at least a few wavelengths long. The aperture is excited by a ***uniform field distribution*** (i.e., constant value) given by

$$\widetilde{E}_{\mathrm{a}}(x_{\mathrm{a}},y_{\mathrm{a}})=\begin{cases}E_0 & \text{for } -l_x/2\le x_{\mathrm{a}}\le l_x/2\\ & \text{and } -l_y/2\le y_{\mathrm{a}}\le l_y/2,\\ 0 & \text{otherwise.}\end{cases} \tag{9.77}$$

To keep the mathematics simple, let us confine our examination to the radiation pattern at a fixed range R in the x–z plane, which corresponds to $\phi=0$. In this case, Eq. (9.75) simplifies to

$$\widetilde{h}(\theta)=\int_{y_{\mathrm{a}}=-l_y/2}^{l_y/2}\int_{x_{\mathrm{a}}=-l_x/2}^{l_x/2}E_0\,\exp[jkx_{\mathrm{a}}\sin\theta]\,dx_{\mathrm{a}}\,dy_{\mathrm{a}}. \tag{9.78}$$

In preparation for performing the integration in Eq. (9.78), we introduce the intermediate variable u defined as

$$u=k\sin\theta=\frac{2\pi\sin\theta}{\lambda}. \tag{9.79}$$

Hence,

$$\begin{aligned}\widetilde{h}(\theta)&=E_0\int_{-l_x/2}^{l_x/2}e^{jux_{\mathrm{a}}}\,dx_{\mathrm{a}}\cdot\int_{-l_y/2}^{l_y/2}dy_{\mathrm{a}}\\ &=E_0\left[\frac{e^{jul_x/2}-e^{-jul_x/2}}{ju}\right]\cdot l_y\\ &=\frac{2E_0l_y}{u}\left[\frac{e^{jul_x/2}-e^{-jul_x/2}}{2j}\right]=\frac{2E_0l_y}{u}\sin(ul_x/2).\end{aligned} \tag{9.80}$$

Upon replacing u with its defining expression, we have

$$\begin{aligned}\widetilde{h}(\theta)&=\frac{2E_0l_y}{\left(\frac{2\pi}{\lambda}\sin\theta\right)}\sin(\pi l_x\sin\theta/\lambda)\\ &=E_0l_xl_y\,\frac{\sin(\pi l_x\sin\theta/\lambda)}{\pi l_x\sin\theta/\lambda}\\ &=E_0A_{\mathrm{p}}\,\mathrm{sinc}(\pi l_x\sin\theta/\lambda),\end{aligned} \tag{9.81}$$

where $A_{\mathrm{p}}=l_xl_y$ is the physical area of the aperture. Also, we used the standard definition of the sinc function, which for any argument t is defined as

$$\mathrm{sinc}\,t=\frac{\sin t}{t}. \tag{9.82}$$

Using Eq. (9.76), we obtain the following expression for the power density at the observation point:

$$S(R,\theta)=S_0\,\mathrm{sinc}^2(\pi l_x\sin\theta/\lambda)\quad(x\text{–}z\text{ plane}), \tag{9.83}$$

where $S_0=E_0^2A_{\mathrm{p}}^2/(2\eta_0\lambda^2R^2)$.

观察点 Q 的位置由孔径中心和点 Q 之间的范围 R 以及角度 θ 和 ϕ（指定。 9-22），共同定义了观察点相对于孔径坐标系的方向。在我们处理偶极子天线时，我们将偶极子沿 z 轴定向，并将 θ 称为天顶角。在本上下文中，z 轴与包含天线孔径的平面正交。另外，θ通常称为仰角。入射到 Q 点的波的电场相量表示为 ˜E(R, θ , ϕ)。基尔霍夫的标量衍射理论提供了辐射场 ˜E(R, θ , ϕ) 与孔径照明 ˜Ea(xa, ya) 之间的关系：

$$\widetilde{E}(R,\theta,\phi)=\frac{j}{\lambda}\left(\frac{e^{-jkR}}{R}\right)\widetilde{h}(\theta,\phi), \tag{9.74}$$

在哪里

$$\widetilde{h}(\theta,\phi)=\iint_{-\infty}^{\infty}\widetilde{E}_{\mathrm{a}}(x_{\mathrm{a}},y_{\mathrm{a}}) \cdot\exp[jk\sin\theta(x_{\mathrm{a}}\cos\phi+y_{\mathrm{a}}\sin\phi)]\,dx_{\mathrm{a}}\,dy_{\mathrm{a}}. \tag{9.75}$$

我们将 ˜h(θ , ϕ) 称为 ˜E(R, θ , ϕ) 的形状因子。它的积分被写成无限极限，且 ˜Ea(xa, ya) 在孔径外同样为零。球面传播因子 (e−jkR/R) 解释了孔径中心和观察点之间的波传播，而 ˜h(θ , ϕ) 表示激发场 ˜Ea(xa, ya) 在孔径范围内的积分，并考虑[通过方程中的指数函数]。 (9.75)]R 和 s 之间距离的近似偏差，其中 s 是到孔径平面中任意点 (xa, ya) 的距离（见图 9-22）。

► 在基尔霍夫标量公式中，辐射场 ˜E(R, θ , ϕ) 的偏振方向与孔径场 ˜Ea(xa, ya) 的偏振方向相同。 ◄

此外，辐射波的功率密度由下式给出

$$S(R,\theta,\phi)=\frac{|\widetilde{E}(R,\theta,\phi)|^2}{2\eta_0}=\frac{|\widetilde{h}(\theta,\phi)|^2}{2\eta_0\lambda^2R^2}. \tag{9.76}$$

9-8 孔径分布均匀的矩形孔径

为了说明标量衍射技术，考虑一个高度为 lx 和宽度为 ly 的矩形孔径，两者都至少是几个波长长。孔径由均匀场分布（即恒定值）激发，由下式给出

$$\widetilde{E}_{\mathrm{a}}(x_{\mathrm{a}},y_{\mathrm{a}})=\begin{cases}E_0 & \text{for } -l_x/2\le x_{\mathrm{a}}\le l_x/2\\ & \text{and } -l_y/2\le y_{\mathrm{a}}\le l_y/2,\\ 0 & \text{otherwise.}\end{cases} \tag{9.77}$$

为了保持数学简单，让我们将我们的检查限制在 x-z 平面中固定范围 R 的辐射方向图上，该范围对应于 ϕ = 0。 (9.75) 简化为

$$\widetilde{h}(\theta)=\int_{y_{\mathrm{a}}=-l_y/2}^{l_y/2}\int_{x_{\mathrm{a}}=-l_x/2}^{l_x/2}E_0\,\exp[jkx_{\mathrm{a}}\sin\theta]\,dx_{\mathrm{a}}\,dy_{\mathrm{a}}. \tag{9.78}$$

准备执行方程式中的积分。 (9.78)，我们引入中间变量 u 定义为

$$u=k\sin\theta=\frac{2\pi\sin\theta}{\lambda}. \tag{9.79}$$

因此，

$$\begin{aligned}\widetilde{h}(\theta)&=E_0\int_{-l_x/2}^{l_x/2}e^{jux_{\mathrm{a}}}\,dx_{\mathrm{a}}\cdot\int_{-l_y/2}^{l_y/2}dy_{\mathrm{a}}\\ &=E_0\left[\frac{e^{jul_x/2}-e^{-jul_x/2}}{ju}\right]\cdot l_y\\ &=\frac{2E_0l_y}{u}\left[\frac{e^{jul_x/2}-e^{-jul_x/2}}{2j}\right]=\frac{2E_0l_y}{u}\sin(ul_x/2).\end{aligned} \tag{9.80}$$

将 u 替换为其定义表达式后，我们有

$$\begin{aligned}\widetilde{h}(\theta)&=\frac{2E_0l_y}{\left(\frac{2\pi}{\lambda}\sin\theta\right)}\sin(\pi l_x\sin\theta/\lambda)\\ &=E_0l_xl_y\,\frac{\sin(\pi l_x\sin\theta/\lambda)}{\pi l_x\sin\theta/\lambda}\\ &=E_0A_{\mathrm{p}}\,\mathrm{sinc}(\pi l_x\sin\theta/\lambda),\end{aligned} \tag{9.81}$$

其中 Ap = lxly 是孔径的物理面积。此外，我们使用了 sin c 函数的标准定义，对于任何参数 t 都定义为

$$\mathrm{sinc}\,t=\frac{\sin t}{t}. \tag{9.82}$$

使用方程式根据方程（9.76），我们得到观测点处功率密度的表达式如下：

$$S(R,\theta)=S_0\,\mathrm{sinc}^2(\pi l_x\sin\theta/\lambda)\quad(x\text{–}z\text{ plane}), \tag{9.83}$$

其中 $S_0=E_0^2\,A_{\mathrm{p}}^2/(2\eta_0\lambda^2R^2)$。